Retail Data Analytics – Submission Report

KPI Dictionary • Data Profiling • Data Quality Contract

1. Dataset Overview

The supplied retail orders CSV was profiled and cleaned using the supplied retail data dictionary. The source file contains 12 rows. One exact duplicate was removed before KPI validation. The final KPI population contains 6 records that pass all mandatory quality rules.

Item	Result
Source rows	12
Exact duplicate rows removed	1
Valid KPI population	6
Net Revenue	■10,060.65

2. KPI Dictionary

KPI	Definition	Formula	Owner	Refresh
Total Valid Orders	Count of valid unique orders	COUNT(DISTINCT order_id)	Retail Operations	Daily
Net Revenue	Revenue after discount	Σ(quantity × price × (1 – discount/100))	Retail Operations	Daily
Average Order Value	Average net revenue per order	Net Revenue / Valid Orders	Retail Operations	Daily
Total Quantity Sold	Total units in valid orders	Σ(quantity)	Retail Operations	Daily
Average Discount %	Average discount percentage	AVG(discount_pct)	Retail Operations	Daily
Paid Orders	Orders with Paid status	COUNT(order_id WHERE Paid)	Finance / Operations	Daily
Pending Orders	Orders with Pending status	COUNT(order_id WHERE Pending)	Finance / Operations	Daily
Refunded Orders	Orders with Refunded status	COUNT(order_id WHERE Refunded)	Finance / Operations	Daily

3. Data Quality Profile

The following issues were identified from the supplied raw file. Records failing mandatory rules were excluded from the KPI population unless a documented normalization/assumption was applied.

Issue	Evidence	Action
Duplicate order	RT-1004 appears as an exact duplicate.	Removed duplicate; retained first occurrence.
Missing city	RT-1005 has blank city.	Excluded from KPI population; source correction required.
Invalid date	RT-1006 contains 2026-13-10.	Excluded; valid date required.
Invalid quantity	RT-1006 contains -1.	Excluded; quantity must be > 0.
Invalid discount	RT-1007 contains 105%.	Excluded; discount must be 0–100%.
Invalid quantity type	RT-1008 contains 'two'.	Excluded; quantity must be a positive integer.
Missing discount	RT-1003 has blank discount.	Treated as 0 under documented exercise assumption.
Missing order date	RT-1011 has no date.	Excluded; order date is required.
Date format inconsistency	RT-1002 uses DD/MM/YYYY.	Normalized to 2026-01-03.
Case inconsistency	Examples include 'paid' / 'student'.	Normalized to standard values.

4. Data Quality Contract

Area	Rule	Failure Threshold	Action
Order ID – Completeness	Required and non-empty	0%	Reject/hold affected record
Order ID – Uniqueness	Unique per order	0%	Deduplicate and investigate
Order Date – Validity	Valid date in expected range	0%	Reject/hold invalid record
Customer Segment	Student/Fresher/Professional	0%	Normalize or reject unknown value
City – Completeness	Required non-empty text	0%	Hold for source correction
Category	Allowed category values only	0%	Reject unknown value
Quantity	Whole number greater than 0	0%	Reject/hold invalid record
Unit Price	Numeric and non-negative	0%	Reject/hold invalid record
Discount	0–100%; missing only with approved rule	0%	Flag and apply approved rule
Payment Status	Paid/Pending/Failed/Refunded	0%	Normalize or reject unknown
Freshness	Refreshed within agreed daily SLA	1 day	Escalate to data owner

5. Submission Note

This report is prepared from the exact CSV and data dictionary supplied in this conversation. The task screenshot mentions a 10,000+ row dataset, but the supplied CSV contains 12 rows. Therefore, this report does not claim that the supplied file contains 10,000+ rows.

Supporting deliverables: KPI Dictionary + Data Profile Excel, executable Python notebook, and Data Quality Contract.